EXPERIMENT NO 1

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Aim: To load a dataset using python and perform data exploration and summarization techniques to understand datasets structure, data types, missing values, and basic statistical measures.

```
In [2]: import pandas as pd

In [3]: import numpy as np

In [4]: df = pd.read_csv("StudentPerformance.csv")
```

```
In [5]: df.head(10)
```

Out[5]:

	Hours Studied	Previous Scores	Extracurricular Activities	Sleep Hours	Sample Question Papers Practiced	Performance Index
0	7	99	Yes	9	1	91.0
1	4	82	No	4	2	65.0
2	8	51	Yes	7	2	45.0
3	5	52	Yes	5	2	36.0
4	7	75	No	8	5	66.0
5	3	78	No	9	6	61.0
6	7	73	Yes	5	6	63.0
7	8	45	Yes	4	6	42.0
8	5	77	No	8	2	61.0
9	4	89	No	4	0	69.0

In [6]:
```
df.tail()
```

Out[6]:

	Hours Studied	Previous Scores	Extracurricular Activities	Sleep Hours	Sample Question Papers Practiced	Performance Index
9995	1	49	Yes	4	2	23.0
9996	7	64	Yes	8	5	58.0
9997	6	83	Yes	8	5	74.0
9998	9	97	Yes	7	0	95.0
9999	7	74	No	8	1	64.0

In [7]:
```
df.tail(5)
```

Out[7]:

	Hours Studied	Previous Scores	Extracurricular Activities	Sleep Hours	Sample Question Papers Practiced	Performance Index
9995	1	49	Yes	4	2	23.0
9996	7	64	Yes	8	5	58.0
9997	6	83	Yes	8	5	74.0
9998	9	97	Yes	7	0	95.0
9999	7	74	No	8	1	64.0

In [8]:
```
df.shape
```

Out[8]:
```
(10000, 6)
```

In [9]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 6 columns):
 #   Column                            Non-Null Count  Dtype
---  ------                            --------------  -----
 0   Hours Studied                     10000 non-null  int64
 1   Previous Scores                   10000 non-null  int64
 2   Extracurricular Activities        10000 non-null  object
 3   Sleep Hours                       10000 non-null  int64
 4   Sample Question Papers Practiced  10000 non-null  int64
 5   Performance Index                 10000 non-null  float64
dtypes: float64(1), int64(4), object(1)
memory usage: 468.9+ KB
```

In [10]:
```
df.columns
```

Out[10]:
```
Index(['Hours Studied', 'Previous Scores', 'Extracurricular Activities',
       'Sleep Hours', 'Sample Question Papers Practiced', 'Performance Index'],
      dtype='object')
```

In [15]:
```
df.isnull()
```

Out[15]:

	Hours Studied	Previous Scores	Extracurricular Activities	Sleep Hours	Sample Question Papers Practiced	Performance Index
0	False	False	False	False	False	False
1	False	False	False	False	False	False
2	False	False	False	False	False	False
3	False	False	False	False	False	False
4	False	False	False	False	False	False
...	...	...	...	...	...	...
9995	False	False	False	False	False	False
9996	False	False	False	False	False	False
9997	False	False	False	False	False	False
9998	False	False	False	False	False	False
9999	False	False	False	False	False	False

10000 rows × 6 columns

In [20]:
```
df.isnull().sum().count
```

Out[20]:
```
<bound method Series.count of Hours Studied
Previous Scores                     0
Extracurricular Activities          0
Sleep Hours                         0
Sample Question Papers Practiced    0
Performance Index                   0
dtype: int64>
```

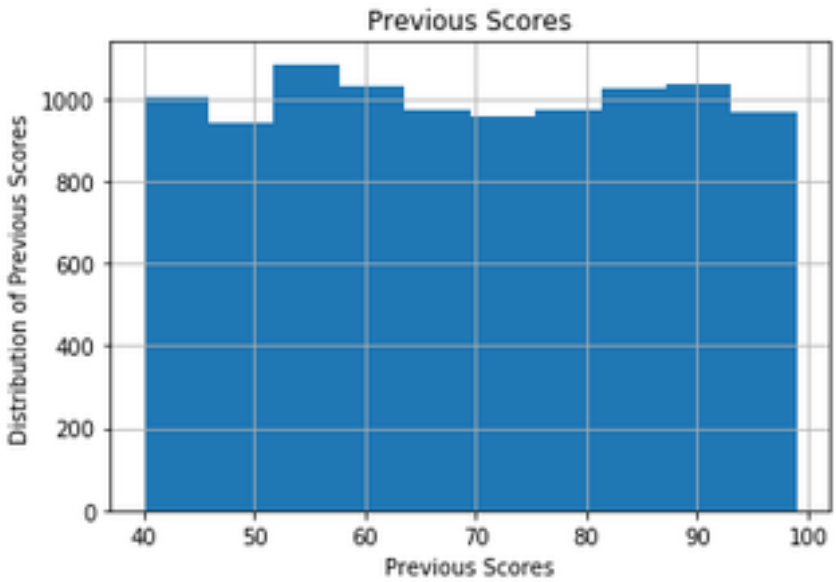
In [13]:
```
df.describe()
```

Out[13]:

	Hours Studied	Previous Scores	Sleep Hours	Sample Question Papers Practiced	Performance Index
count	10000.000000	10000.000000	10000.000000	10000.000000	10000.000000
mean	4.992900	69.445700	6.530600	4.583300	55.224800
std	2.589309	17.343152	1.695863	2.867348	19.212558
min	1.000000	40.000000	4.000000	0.000000	10.000000
25%	3.000000	54.000000	5.000000	2.000000	40.000000
50%	5.000000	69.000000	7.000000	5.000000	55.000000
75%	7.000000	85.000000	8.000000	7.000000	71.000000
max	9.000000	99.000000	9.000000	9.000000	100.000000

In [21]:
```
import matplotlib.pyplot as plt
```

In [25]:
```
df['Previous Scores'].hist()
plt.xlabel("Previous Scores")
plt.ylabel("Distribution of Previous Scores")
plt.title("Previous Scores")
plt.show()
```



In [26]:
```
df['Sleep Hours'].hist()
plt.xlabel("Sleep Hours")
plt.ylabel("Distribution of Sleep Hours")
plt.title("Sleep Hours")
plt.show()
```